MATHS MANTHAN
• TUITION CLASSES •
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Subject: Mathematics
Class: Class X (CBSE)

CBSE CLASS 10 – WEEKLY TEST 02
Chapter 1: Real Numbers

Time: 45 Minutes
MM: 20

Document Type: Official Step-by-Step Marking Scheme **Published for:** mathsmanthan.co.in

CBSE STEP-MARKING GUIDELINES & EVALUATION PROTOCOL:

1. Full marks require explicit theorem/formula declaration, complete prime factorisation or contradiction steps, and boxed final answers with units.
2. In Assertion–Reason questions, truth values of both (A) and (R) as well as whether (R) validly explains (A) are explicitly established.
3. Partial marks are allocated per step in brackets: **[0.5M, 1M, 2M]**. Alternative valid methods receive full credit.

Official Evaluation Key | Maths Manthan Meerut

SECTION A: MCQs & Assertion–Reason

MATHS MANTHAN | (5 × 1 = 5 Marks)

Q1. If two positive integers p and q can be expressed as $p = ab^2$ and $q = a^3b$, where a, b are prime numbers, then LCM(p, q) is: [1]

(a) ab (b) a^2b^2 (c) a^3b^2 (d) a^4b^3

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (c) a^3b^2

Given $p = a^1b^2$ and $q = a^3b^1$. We know:

Formula Used: LCM(p, q) = Product of greatest powers of all prime factors

Greatest power of a is $\max(1, 3) = 3$, and of b is $\max(2, 1) = 2$. Thus LCM$(p, q) = \boxed{a^3b^2}$. [1M]

Q2. The product of a non-zero rational number and an irrational number is: [1]

(a) always rational (b) always irrational
(c) rational or irrational (d) one

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (b) always irrational

Let $r = \frac{p}{q} \neq 0$ be rational and x be irrational. We know by contradiction:

Formula Used: $r \neq 0 \in \mathbb{Q},\ x \notin \mathbb{Q} \implies r \cdot x \notin \mathbb{Q}$

If $r \cdot x = s \in \mathbb{Q}$, then $x = \frac{s}{r} \in \mathbb{Q}$ (since $r \neq 0$), a contradiction. Hence the product is $\boxed{\text{always irrational}}$. [1M]

Q3. The least number that is divisible by all the natural numbers from 1 to 10 (both inclusive) is: [1]

(a) 10 (b) 100 (c) 504 (d) 2520

▶ STEP-BY-STEP SOLUTION: [1 Mark]

Correct Option: (d) 2520

Factorising 1 to 10: $2, 3, 2^2, 5, 2 \cdot 3, 7, 2^3, 3^2, 2 \cdot 5$. We know:

Formula Used: Least common multiple = LCM$(1, 2, 3, \ldots, 10)$

LCM $= 2^3 \times 3^2 \times 5 \times 7 = 8 \times 9 \times 5 \times 7 = 72 \times 35 = \boxed{2520}$. [1M]

Q4. If HCF$(26, 169) = 13$, then LCM$(26, 169)$ is: [1]

(a) 26 (b) 52 (c) 338 (d) 13

▶ STEP-BY-STEP SOLUTION: [1 Mark]

Correct Option: (c) 338

Given HCF$(26, 169) = 13$. We know the two-number identity:

Formula Used: HCF$(a, b) \times$ LCM$(a, b) = a \times b$

$$\text{LCM}(26, 169) = \frac{26 \times 169}{13} = 2 \times 169 = \boxed{338}.$$ [1M]

Q5. Assertion (A): The HCF of two numbers is 5 and their product is 150, then their LCM is 30.
Reason (R): For any two positive integers a and b, HCF$(a, b) \times$ LCM$(a, b) = a \times b$. [1]

(A) Both (A) and (R) are true and (R) is the correct explanation of (A).
(B) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(C) (A) is true but (R) is false. (D) (A) is false but (R) is true.

▶ STEP-BY-STEP SOLUTION: [1 Mark]

Correct Option: (A) Both (A) and (R) are true and (R) is the correct explanation of (A).

We know the product-HCF-LCM identity:

Formula Used: HCF$(a, b) \times$ LCM$(a, b) = a \times b$

Reason (R) is true by theorem. Substituting HCF $= 5$ and $a \times b = 150$: LCM $= \frac{150}{5} = 30$, confirming Assertion (A). Thus (R) directly explains (A). [1M]

Q6. Check whether 6^n can end with the digit 0 for any natural number $n \in \mathbb{N}$. [2]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

We know the divisibility condition for a base-10 trailing zero via the Fundamental Theorem of Arithmetic:

Formula Used: A number ends with digit 0 $\iff$ $10 = 2 \times 5$ divides it (both primes 2 and 5 occur)

Prime factorising 6^n for any $n \in \mathbb{N}$:

$$6^n = (2 \times 3)^n = 2^n \times 3^n. \qquad [1M]$$

The only primes in 6^n are 2 and 3; prime 5 is absent. By the uniqueness of prime factorisation (FTA), 5 can never divide 6^n. Therefore, $\boxed{6^n \text{ can NEVER end with digit 0 for any } n \in \mathbb{N}.}$ [1M]

Q7. Prove that $\sqrt{7}$ is an irrational number. [3]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]

Assumption: Suppose, on the contrary, that $\sqrt{7}$ is rational. Then there exist integers a and b ($b \neq 0$) with $\text{HCF}(a, b) = 1$ (co-prime) such that:

$$\sqrt{7} = \frac{a}{b} \implies a = b\sqrt{7}.$$

Squaring both sides:

$$a^2 = 7b^2. \qquad \text{— (1)} \qquad [1M]$$

We know the Fundamental Theorem of Arithmetic divisibility theorem:

Formula Used: If a prime p divides a^2, then p also divides a.

From (1), 7 divides a^2; since 7 is prime, 7 divides a. Hence $a = 7c$ for some integer c. [1M]
Substituting $a = 7c$ back into (1):

$$(7c)^2 = 7b^2 \implies 49c^2 = 7b^2 \implies b^2 = 7c^2.$$

Thus 7 divides b^2, so 7 also divides b. Therefore 7 is a common factor of both a and b, contradicting our assumption that a and b are co-prime ($\text{HCF}(a, b) = 1$).

This contradiction arose from assuming $\sqrt{7}$ is rational. Hence $\boxed{\sqrt{7} \text{ is an irrational number.}}$ [1M]

Q8. Prove that $2\sqrt{3}-1$ is an irrational number. **[3]**

▶ STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 3 Marks]

Suppose, to the contrary, that $2\sqrt{3}-1$ is rational. Then there exist co-prime integers a, b $(b \neq 0)$ such that:

$$2\sqrt{3}-1=\frac{a}{b} \implies 2\sqrt{3}=\frac{a}{b}+1=\frac{a+b}{b} \implies \sqrt{3}=\frac{a+b}{2b}. \qquad [1M]$$

We know the closure property of rational numbers and prime-root irrationality:

Formula Used: For integers $p, q \neq 0$, $\frac{p}{q} \in \mathbb{Q}$; but $\sqrt{p}$ is irrational for prime p

Since a, b are integers with $b \neq 0$, both $a+b$ and $2b \neq 0$ are integers, making $\frac{a+b}{2b}$ rational, which forces $\sqrt{3}$ to be rational. [1M]

Verification that $\sqrt{3}$ is irrational: If $\sqrt{3}=\frac{u}{v}$ with $\text{HCF}(u,v)=1$, then $u^2=3v^2 \implies 3 \mid u \implies u=3k \implies v^2=3k^2 \implies 3 \mid v$, contradicting co-primality. Hence $\sqrt{3}$ is irrational. Thus our assumption is false, so $\boxed{2\sqrt{3}-1 \text{ is irrational.}}$ [1M]

Q9. Prove that $\sqrt{2}+\sqrt{3}$ is an irrational number. **[3]**

▶ STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 3 Marks]

Suppose, on the contrary, that $r=\sqrt{2}+\sqrt{3}$ is rational $(r \in \mathbb{Q},\ r>0)$. Isolating $\sqrt{3}=r-\sqrt{2}$ and squaring both sides:

$$(\sqrt{3})^2=(r-\sqrt{2})^2 \implies 3=r^2+2-2r\sqrt{2}. \qquad [1M]$$

Rearranging to isolate $\sqrt{2}$:

$$2r\sqrt{2}=r^2+2-3=r^2-1 \implies \sqrt{2}=\frac{r^2-1}{2r}.$$

We know rational numbers are closed under addition, subtraction, multiplication, and non-zero division:

Formula Used: $r \in \mathbb{Q} \setminus \{0\} \implies \frac{r^2-1}{2r} \in \mathbb{Q}$

Since r is rational and $r \neq 0$, $\frac{r^2-1}{2r}$ is rational, forcing $\sqrt{2}$ to be rational. [1M]

However, if $\sqrt{2}=\frac{a}{b}$ with $\text{HCF}(a,b)=1$, then $a^2=2b^2 \implies 2 \mid a$ and $2 \mid b$, contradicting $\text{HCF}(a,b)=1$, so $\sqrt{2}$ is irrational. This contradiction proves $\boxed{\sqrt{2}+\sqrt{3} \text{ is irrational.}}$ [1M]

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Q10. CASE STUDY 02: NATIONAL EDUCATIONAL SEMINAR ROOM ALLOCATION [4 Marks]

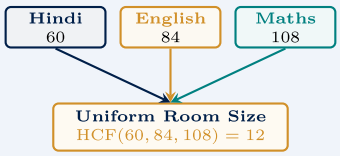
The Ministry of Education is organising a 3-day National Seminar on Mathematical Sciences in New Delhi. The participating teachers are registered across three disciplines: Hindi (60 participants), English (84 participants), and Mathematics (108 participants). The event coordinator must allocate participants into parallel breakout rooms such that in each room, exactly the same number of participants are seated, and all participants in any given room belong to the same discipline.

(i) Find the maximum number of participants that can be accommodated in each breakout room. [2]

(ii) Calculate the minimum total number of breakout rooms required for the seminar. [1]

(iii) Express the Mathematics participant count (108) in canonical prime factorisation form. [1]

▶ CASE-BASED SOLUTION & MARKING SCHEME: [Total: 4 Marks]



Allocation Principle: To minimize the number of rooms under single-discipline uniform seating, room capacity must be the largest common divisor of 60, 84, and 108, which is 12.

We know the HCF formula via prime factorisation:

Formula Used: HCF(a, b, c) = Product of smallest powers of common prime factors

(i) Max capacity per room: Prime factorising 60, 84, and 108:

$$60 = 2^2 \times 3^1 \times 5, \qquad 84 = 2^2 \times 3^1 \times 7, \qquad 108 = 2^2 \times 3^3.$$

Common primes are 2 and 3, with smallest powers 2^2 and 3^1:

$$\text{HCF}(60, 84, 108) = 2^2 \times 3 = 4 \times 3 = \boxed{\text{12 participants per room}}. \qquad [2M]$$

(ii) Minimum total rooms required:

$$\text{Rooms} = \frac{60}{12} + \frac{84}{12} + \frac{108}{12} = 5 + 7 + 9 = \boxed{\text{21 breakout rooms}}. \qquad [1M]$$

(iii) Canonical prime factorisation of 108: $108 = 4 \times 27 = \boxed{2^2 \times 3^3}$. [1M]

★★★ END OF SOLUTIONS · MATHS MANTHAN MEERUT ★★★